

# ARASH JALIL KHABBAZI

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## SUMMARY

I'm a PhD candidate in Mechanical Engineering at Purdue University, working on predictive control and reinforcement learning for building energy systems. My work draws on machine learning, control theory, modeling and simulation, and field demonstrations in real buildings, applied to HVAC control, measurement and verification, and other distributed energy resources projects. I want to build a research career in scientific machine learning and control, pairing rigorous methods with real-world testing to make control systems more reliable and easier to deploy at scale.

## EDUCATION

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|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| <b>Purdue University, West Lafayette, IN</b><br>PhD in Mechanical Engineering<br>Specialization in Computational Science and Engineering                       | 08/2023 – Present<br><b>GPA: 3.9/4.0</b>                           |
| <b>University of British Columbia, BC, Canada</b><br>MS in Mechanical Engineering<br>Thesis: “Mixing Gaseous Hydrogen into Natural Gas Distribution Pipelines” | 05/2021 – 08/2023<br><b>GPA: 94% (4.0/4.0)</b>                     |
| <b>University of Tabriz, EA, Iran</b><br>BS in Mechanical Engineering                                                                                          | 08/2016 – 07/2020<br><b>GPA: 19.12/20 (4.0/4.0) – Rank: 1/155+</b> |

## SELECT AWARDS AND RECOGNITION

|                                                                                         |      |
|-----------------------------------------------------------------------------------------|------|
| <b>ASHRAE Graduate Student Grant-in-Aid Award</b>                                       | 2025 |
| <b>Best PhD Forum Presentation Runner-up Award, ACM BuildSys</b>                        | 2025 |
| <b>Best Paper Award, CSME International Congress</b>                                    | 2023 |
| <b>Best Presentation Award, CSME International Congress (Advanced Energy Symposium)</b> | 2022 |
| <b>UBC Graduate Scholarship, University of British Columbia</b>                         | 2022 |
| <b>UBC Dean’s Entrance Scholarship, University of British Columbia</b>                  | 2021 |

## PUBLICATIONS

### Journal Articles

2. **A.J. Khabbazi**, E.N. Pergantis, L.D. Reyes Premer, P. Papageorgiou, A.H. Lee, J.E. Braun, G.P. Henze, and K.J. Kircher, “Lessons learned from field demonstrations of model predictive control and reinforcement learning for residential and commercial HVAC: A review.” *Applied Energy*, 2025. 🔗
1. **A.J. Khabbazi**, M. Zabihi, R. Li, M. Hill, V. Chou, and J. Quinn, “Mixing hydrogen into natural gas distribution pipeline system through Tee junctions.” *International Journal of Hydrogen Energy*, 2024. 🔗

### Preprints

1. L.D. Reyes Premer, **A.J. Khabbazi**, and K.J. Kircher, “Indirect Data-Driven Predictive Control and the State-Space Predictor.” *arXiv:2602.10936*, 2026. *Under review*.

### Conference Papers

6. W.G. Dierking, **A.J. Khabbazi**, L.D. Reyes Premer, and K.J. Kircher, “Scalable Supervisory HVAC Control for Linear Objectives.” *65th IEEE Conference on Decision and Control (CDC)*, 2026.
5. L.D. Reyes Premer, E.N. Pergantis, **A.J. Khabbazi**, F. Wu, and K.J. Kircher, “Evaluating Power Control Strategies for Avoiding Residential Panel Upgrades in Cold Climates.” *ACEEE Summer Study on Energy Efficiency in Buildings*, 2026.
4. **A.J. Khabbazi** and K.J. Kircher, “Small HVAC Control Demonstrations in Larger Buildings Often Overestimate Savings.” *American Control Conference (ACC)*, 2026.
3. E.N. Pergantis, L.D. Reyes Premer, **A.J. Khabbazi**, Priyadarshan, F. Wu, D. Ziviani, and K.J. Kircher, “Active current limiting control of residential appliances for breaker panel protection across the US: a preliminary study.” *CISBAT*, 2025.
2. **A.J. Khabbazi**, E.N. Pergantis, L.D. Reyes Premer, A.H. Lee, J. Ma, H. Liu, G.P. Henze, and K.J. Kircher, “What Have We Learned From Field Demonstrations of Advanced Commercial HVAC Control?” *8th International High Performance Buildings Conference*, 2024.

1. **A.J. Khabbazi**, M. Zabihi, R. Li, V. Chou, and J. Quinn, “Blending of Hydrogen into a Natural Gas Distribution Pipeline in British Columbia through a Tee Junction for Reducing GHG Emissions.” *Canadian Society for Mechanical Engineering (CSME) International Congress, 2023. Best Paper Award.*

## Abstracts

3. **A.J. Khabbazi**, “PhD Forum Abstract: Field Demonstration of Advanced Commercial HVAC Control for Scalable Deployment.” *12th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys), 2025. Best PhD Forum Presentation Runner-up Award.*
2. **A.J. Khabbazi**, R. Li, and J. Quinn, “Green Hydrogen Supply to Urban Infrastructure and Buildings through Blending into the Existing Grid.” *Canadian Society for Mechanical Engineering (CSME) International Congress, 2022. Best Presentation Award.*
1. **A.J. Khabbazi**, R. Li, J. Quinn, and M. Hoorfar, “The Blending and Transmission of Hydrogen and Natural Gas in Transmission and Distribution Pipelines.” *13th International Green Energy Conference (IGEC-XIII), 2021.*

## TALKS

8. “Small HVAC Control Demonstrations in Larger Buildings Often Overestimate Savings.” *American Control Conference (ACC), New Orleans, LA, 2026.*
7. “Field Demonstration of Advanced Commercial HVAC Control for Scalable Deployment.” *PhD Forum, 12th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys), Golden, CO, 2025.*
6. “What Have We Learned From Field Demonstrations of Advanced Commercial HVAC Control?” *8th International High Performance Buildings Conference, Purdue University, West Lafayette, IN, 2024.*
5. “Field Demonstrations of Advanced Commercial HVAC Control.” *Intelligent Building Operations Workshop, Purdue University, West Lafayette, IN, 2024.*
4. “Mixing Gaseous Hydrogen into Natural Gas Distribution Pipelines.” *Herrick Graduate Seminar, Purdue University, West Lafayette, IN, 2023.*
3. “Blending of Hydrogen into a Natural Gas Distribution Pipeline in British Columbia through a Tee Junction for Reducing GHG Emissions.” *CSME International Congress, Université de Sherbrooke, QC, Canada, 2023.*
2. “Green Hydrogen Supply to Urban Infrastructure and Buildings through Blending into the Existing Grid.” *CSME International Congress, University of Alberta, AB, Canada, 2022.*
1. “The Blending and Transmission of Hydrogen and Natural Gas in Transmission and Distribution Pipelines.” *13th International Green Energy Conference (IGEC-XIII), 2021.*

## LEADERSHIP & MENTORSHIP

- Research Mentor**, Purdue University 2024–Present  
– Mentored and collaborated with three master’s students and one undergraduate researcher on research planning and technical writing, co-authoring publications together.
- Disability Resource Centre Invigilator**, University of British Columbia 2021–2023  
– Supported students receiving academic accommodations during examinations.

## INDUSTRY EXPERIENCE

- Purdue Energy & Utilities**, Graduate Researcher, Energy Systems, *West Lafayette, IN* 02/2026–Present  
– Leading data pipeline and modeling for a pilot that pre-cools buildings to shift campus cooling demand off-peak.  
– Quantified chiller-plant efficiency savings (~\$41k/yr) via a weather-normalized M&V study of air/dirt separators.  
– Analyzed campus building EUI and CHW peak demand before/after renovations for E&U renovation planning.
- Tesla, Inc.**, Intern, Machine Learning for Drive Unit Quality, *Sparks, NV* 05/2025–08/2025  
– Built supervised ML models (XGBoost, DNNs, SVR, and ensembles) for defect detection and quality assessment.  
– Developed data pipelines in MySQL and engineered features to extract high-signal inputs for ML models.  
– Helped deploy a thermal imaging system for non-destructive inspection in drive-unit manufacturing.  
– Delivered ML-driven quality improvements projected to reduce testing costs by over \$1M/year.
- FortisBC**, Graduate Researcher, Renewable Gas, *Vancouver, BC* 05/2021–08/2023  
– Collaborated with FortisBC on an NSERC-funded UBC-FortisBC hydrogen-blending project.  
– Performed CFD simulations and developed ideal- and real-gas mixture models in C.   
– Participated in industry meetings and technical discussions on practical challenges of hydrogen integration.  
– Delivered research resulting in one journal paper, three conference papers, an MS thesis, and two awards.

## TEACHING EXPERIENCE

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**Purdue University**, Graduate Teaching Assistant (PhD)

- ME597: Distributed Energy Resources (DERs) *Spring'26*
  - Supported instruction on DER modeling and control for batteries, solar PV, EVs, and HVAC/thermal storage.
  - Assisted with homework, coding modules, and office hours covering DER concepts.
  - Guided semester projects on DER topics, including virtual power plants and grid integration.

**University of British Columbia**, Graduate Teaching Assistant (MS)

- Engineering Analysis I *Fall'21, Spring'22, Fall'22*
- Fluid Mechanics II Lab *Spring'22, Spring'23*

## ACADEMIC SERVICE

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**Conference and workshop organizing:**

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|-----------|--|----------------------------------------------------------------------------|
| July 2026 |  | Intelligent Building Operations Workshop student chair, West Lafayette, IN |
| July 2024 |  | Intelligent Building Operations Workshop student chair, West Lafayette, IN |

**Society memberships:** ASHRAE, ACM, IEEE, CSME

**Reviewing:** IEEE Transactions on Control Systems Technology; Building and Environment; IEEE Conference on Decision and Control; Herrick Conferences; ACEEE Summer Study on Energy Efficiency in Buildings

## SELECTED COURSES

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|-------------------------------------------------|-----------------------------------------------|
| • ECE570: Artificial Intelligence               | • STAT511: Statistical Methods                |
| • IE690: Reinforcement Learning & Control (RLC) | • ME597: Industrial Internet of Things (IIoT) |
| • AAE568: Applied Optimal Control & Estimation  | • ME597: Distributed Energy Resources (DERs)  |
| • AAE 561: Intro to Convex Optimization         | • ME511: Analysis of Thermal Systems          |

## SKILLS

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**Expertise:** Predictive Control, Reinforcement Learning, Data Structures & Algorithms, Convex Optimization, Control Theory, Machine Learning, Deep Learning, Generative AI, Industrial IoT, Buildings & HVAC, Thermodynamics.

**Programming & Tools:** Python, MATLAB, C, Git, Linux, Docker.

**ML & Optimization:** PyTorch, TensorFlow, scikit-learn, XGBoost, Stable-Baselines3, CVX, CVXPY, Gurobi

**Data Analysis & Visualization:** Pandas, NumPy, SciPy, Matplotlib, Seaborn, Grafana.

**Simulation & Testbeds:** Gymnasium, OCHRE, BOPTEST.